

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 8: Databases

Subtopic 8.1: Database Concepts

Concept 8.1.3: Entity-Relationship (E-R) Diagrams

Total Questions: 8

Mark Schemes begin on Page 8

Question 1 | Source: 9618_w25_qp_11 (Q2.a)

- 2 A relational database, `WORKEXP`ERIENCE, stores data about students and the companies where they complete work experience.

Students can complete multiple work experience placements but can only complete one placement at a time.

Students can complete more than one placement at the same company.

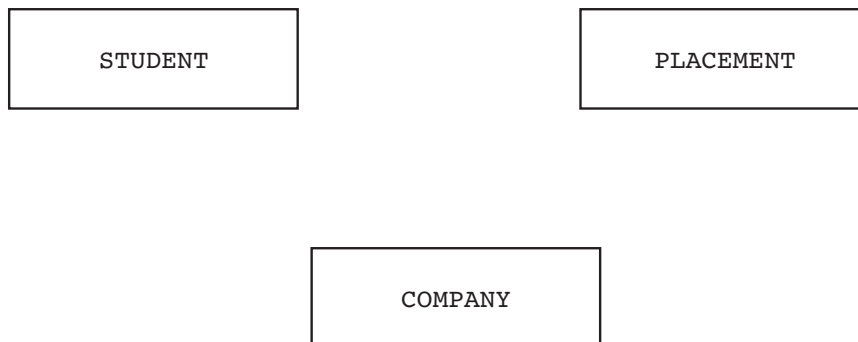
Part of the database is shown:

```
STUDENT(StudentID, FirstName, TelephoneNumber, UniversityName)
```

```
PLACEMENT(PlacementID, StudentID, CompanyID, StartDate, EndDate,  
          Complete)
```

```
COMPANY(CompanyID, CompanyName, MaxStudentsPerPlacement)
```

- (a) Complete the entity-relationship (E-R) diagram for the database `WORKEXP`ERIENCE.



[2]

Question 3 | Source: 9618_s25_qp_11 (Q5.a)

- 5 A company builds and sells furniture to customers. The company stores data about customers, their payment cards and their furniture orders in a database.

The database, `FURNITURE`, has the following tables:

```
CUSTOMER(CustomerID, Name, Phone)
```

```
CUSTOMER_CARD_DATA(CardID, CustomerID, CardType, CardNumber, EndDate)
```

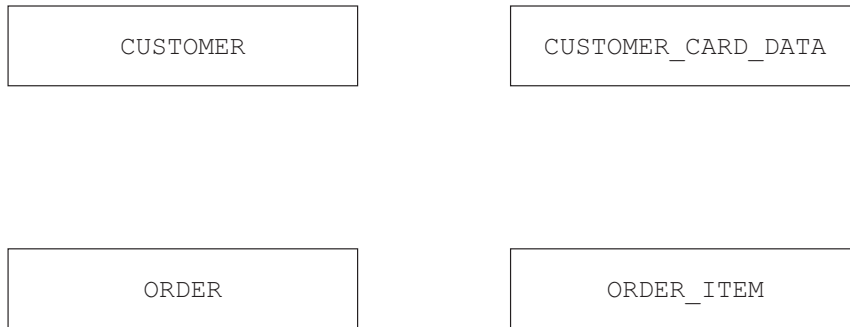
```
ORDER(OrderID, CustomerID, TotalCost, Paid, OrderDate, Complete)
```

```
ORDER_ITEM(OrderItemID, OrderID, Type, Height, Width, Depth, Details)
```

The primary keys are underlined in each table.

The attribute `Complete` in the table `ORDER` stores the Boolean value `TRUE` if the order has been built and `FALSE` if the order has not been built.

(a) Complete the entity-relationship (E-R) diagram for the database.



[3]

Question 4 | Source: 9618_s25_qp_13 (Q6.a)

- 6 A shop sells pens to customers. Customers place an order with the shop and collect the items the next day. The shop uses a database to store the information about the orders.

The database contains the following tables:

CUSTOMER(CustomerID, CustomerName, Email)

ORDER(OrderID, CustomerID, Date, Collected)

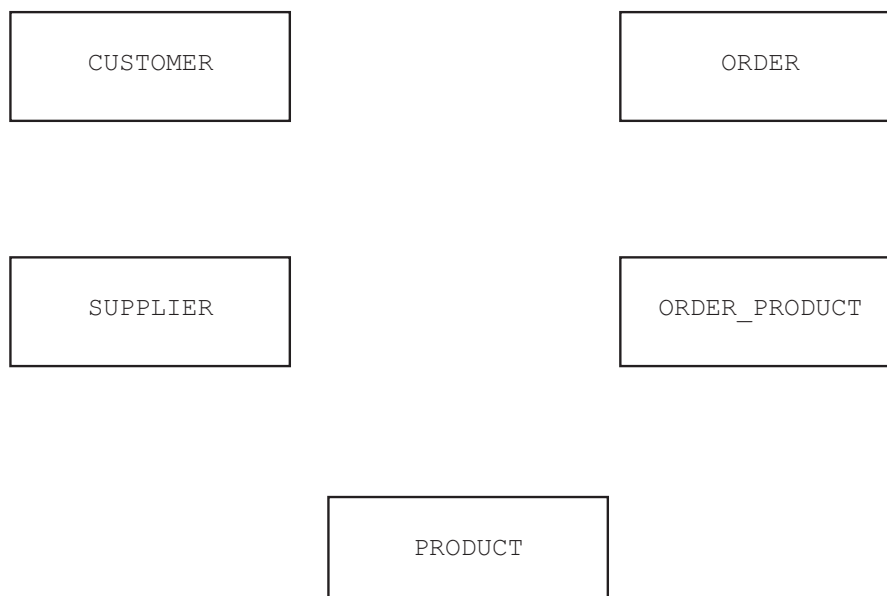
ORDER_PRODUCT(OrderID, ProductID, Quantity)

PRODUCT(ProductID, ProductName, QuantityInBox, Cost, SupplierID)

SUPPLIER(SupplierID, SupplierName, SupplierEmail)

The primary keys are underlined.

(a) Complete the entity-relationship (E-R) diagram for the database.



[4]

Question 5 | Source: 9618_w24_qp_11 (Q2.b.i)

- 2 A shop repairs electronic devices, for example mobile phones and tablet computers. The shop owner stores the data about the repairs using a file-based approach.

(b) The shop owner creates a relational database called `FIXIT`.

The database stores data about the customers and the devices for repair.

Some devices need new parts that are ordered from suppliers.

The database `FIXIT` is designed to include the following tables:

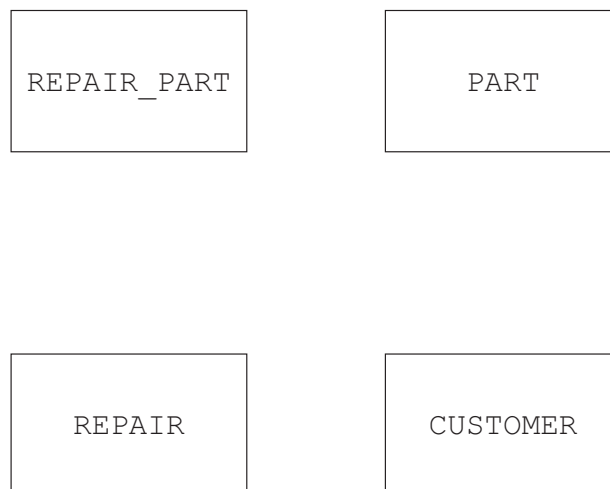
`PART`(PartID, Description, Price, SupplierID)

`CUSTOMER`(CustomerID, FirstName, LastName, ContactNumber)

`REPAIR`(RepairNumber, StartDate, EndDate, CustomerID, Device)

`REPAIR_PART`(PartID, RepairNumber, Quantity)

(i) Complete the entity-relationship (E-R) diagram for the given tables.



[3]

Question 6 | Source: 9618_w23_qp_11 (Q3.a)

- 3 A shop manager has designed a relational database to store customer orders.

The database will have the following tables:

`CUSTOMER`(CustomerID, FirstName, LastName, Town)

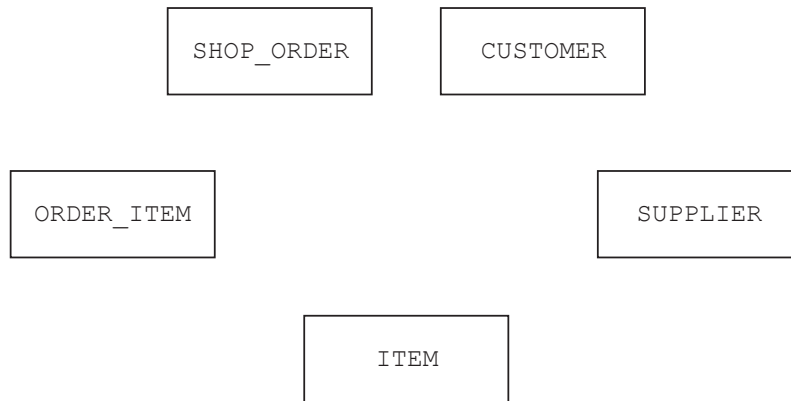
`SHOP_ORDER`(OrderNo, CustomerID, OrderDate)

`SUPPLIER`(SupplierID, EmailAddress, TelephoneNumber)

`ITEM`(ItemNumber, SupplierID, Description, Price)

`ORDER_ITEM`(ItemNumber, OrderNo, Quantity)

(a) Complete the entity-relationship (E-R) diagram for the relational database.



[3]

Question 7 | Source: 9618_w23_qp_13 (Q3.b.i)

- 3 A company sells online Computer Science courses to students in different countries.
The courses are stored on a public cloud.

(b) The company uses a database, *COURSES*, to store data about the courses and their tutors.

Each course starts at different times of the year and may have a different tutor.

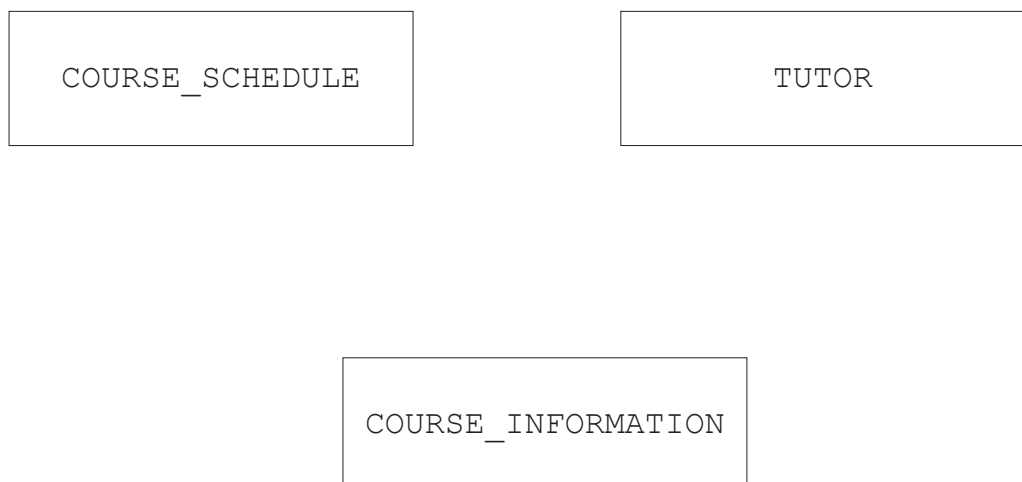
The database has the following structure:

COURSE_INFORMATION(CourseID, Description, Cost)

TUTOR(TutorID, TelephoneNumber, EmailAddress, TutorName)

COURSE_SCHEDULE(CourseID, DateStarted, TutorID)

(i) Complete the entity-relationship (E-R) diagram for the database *COURSES*.



[1]

Question 8 | Source: 9618_w22_qp_11 (Q4.a)

- 4 A photographer creates a relational database to store data about photographs taken at birthday parties.

The database, `PHOTOGRAPHS`, stores details of the customer, the party, the photographs taken and the cameras used.

The photographer has several cameras that are used for taking the photographs at the parties.

Each camera has a specific lens type (for example, XY32Z) and lighting type (for example, F1672).

Data about each photograph is stored in the database including the party at which it was taken, the time it was taken and the camera used.

The database has these four tables:

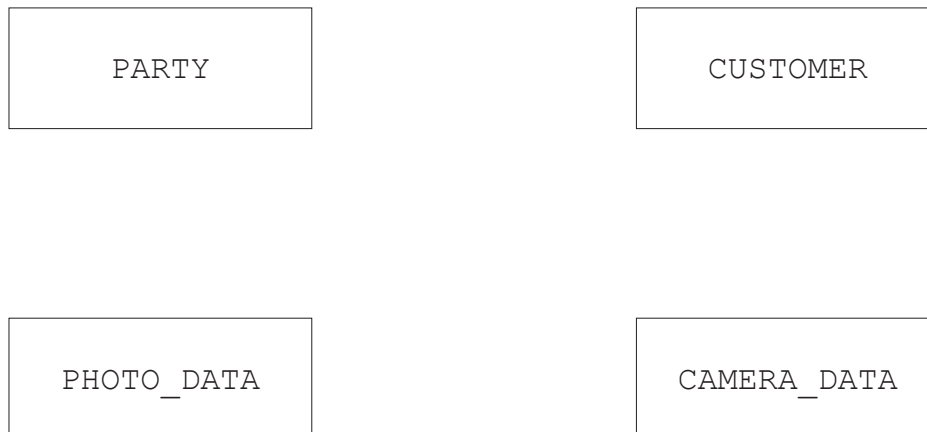
```
CUSTOMER(CustomerID, FirstName, LastName, Telephone)
```

```
PARTY(PartyID, CustomerID, PartyDate, StartTime)
```

```
PHOTO_DATA(PhotoID, PartyID, TimeTaken, CameraID)
```

```
CAMERA_DATA(CameraID, LensType, LightingType)
```

- (a) Complete the entity-relationship (E-R) diagram for the database `PHOTOGRAPHS`.



[3]

Question 9 | Source: 9618_w21_qp_12 (Q6.a.ii)

- 6 A shop sells plants to customers. The shop manager has a relational database to keep track of the sales.

The database, `PLANTSALES`, has the following structure:

```
PLANT(PlantName, QuantityInStock, Cost)
```

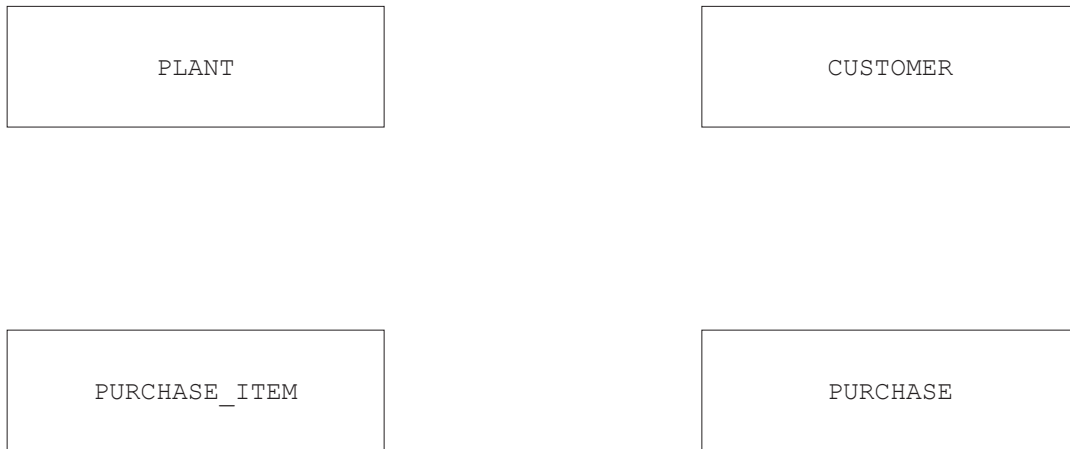
```
CUSTOMER(CustomerID, FirstName, LastName, Address, Email)
```

```
PURCHASE(PurchaseID, CustomerID)
```

```
PURCHASE_ITEM(PurchaseID, PlantName, Quantity)
```

(a) The database is normalised.

(ii) Draw an entity-relationship (E-R) diagram for the database `PLANTSALES`.



[3]

MARK SCHEMES

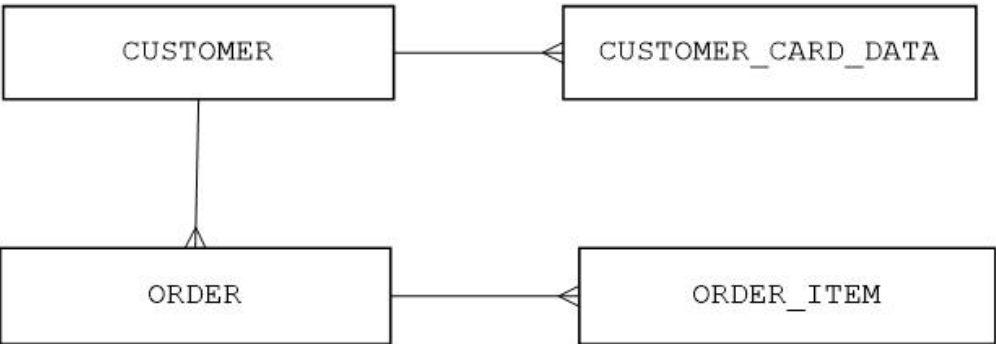
Mark Scheme for Question 1 | Source: 9618_w25_ms_11 (Q2.a)

2(a)	1 mark for each correct relationship, max 2 marks	2
<pre> classDiagram class STUDENT class PLACEMENT class COMPANY STUDENT --> PLACEMENT PLACEMENT --> COMPANY PLACEMENT --> PLACEMENT </pre>		

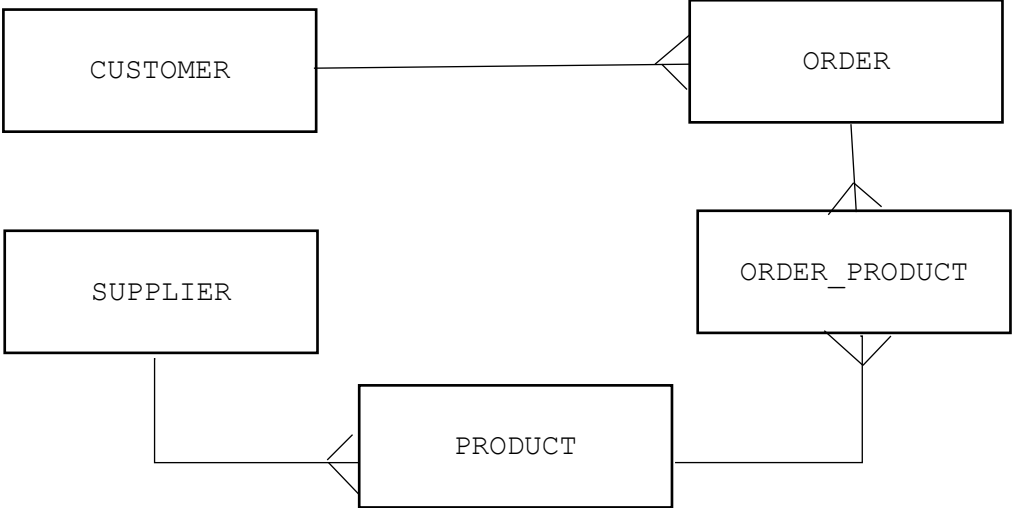
Mark Scheme for Question 2 | Source: 9618_w25_ms_13 (Q5.a)

5(a)	1 mark for each correct relationship, max 3 marks	3
<pre> classDiagram class CUSTOMER class PRODUCT class COMPLAINT class STAFF CUSTOMER --> COMPLAINT PRODUCT --> COMPLAINT STAFF --> COMPLAINT COMPLAINT --> COMPLAINT </pre>		

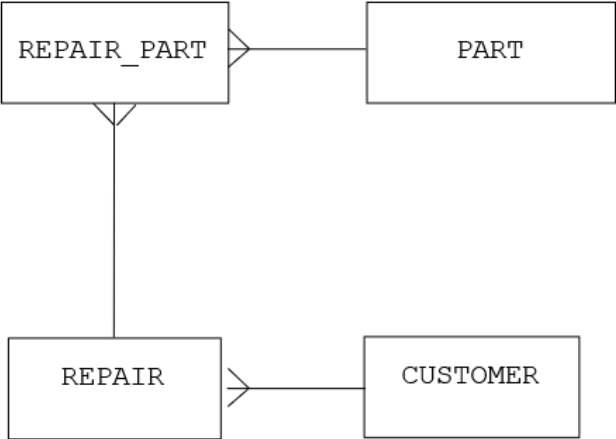
Mark Scheme for Question 3 | Source: 9618_s25_ms_11 (Q5.a)

5(a)	1 mark for each relationship. Max 2 if any extra	3
		

Mark Scheme for Question 4 | Source: 9618_s25_ms_13 (Q6.a)

6(a)	1 mark for each correct relationship	4
		

Mark Scheme for Question 5 | Source: 9618_w24_ms_11 (Q2.b.i)

2(b)(i)	1 mark for each correct relation	3
		

Mark Scheme for Question 6 | Source: 9618_w23_ms_11 (Q3.a)

3(a)	1 mark for each correct relationship or relationships • 1:M between CUSTOMER and SHOP_ORDER • 1:M between SUPPLIER and ITEM • 1:M between SHOP_ORDER and ORDER_ITEM and M:1 between ORDER_ITEM and ITEM <pre>graph LR; SHOP_ORDER --> CUSTOMER</pre>	3
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Mark Scheme for Question 7 | Source: 9618_w23_ms_13 (Q3.b.i)

3(b)(i)	1 mark for both 1:M relationships as follows: <pre>graph TD; COURSE_SCHEDULE --> TUTOR; COURSE_SCHEDULE --> COURSE_INFORMATION</pre>	1
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Mark Scheme for Question 8 | Source: 9618_w22_ms_11 (Q4.a)

4(a)	1 mark for each correct relationship: • 1:M between CUSTOMER and PARTY • 1:M between PARTY and PHOTO_DATA • 1:M between CAMERA_DATA and PHOTO_DATA <pre>graph TD; PARTY --> CUSTOMER; PARTY --> PHOTO_DATA; PHOTO_DATA --> CAMERA_DATA</pre>	3
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6(a)(ii)	1 mark for each correct relationship <pre>classDiagram class PLANT class CUSTOMER class PURCHASE_ITEM class PURCHASE PLANT -- > PURCHASE_ITEM CUSTOMER -- > PURCHASE PURCHASE_ITEM --> PURCHASE</pre> The diagram shows four classes: PLANT, CUSTOMER, PURCHASE_ITEM, and PURCHASE. PLANT and CUSTOMER are generalizations of PURCHASE_ITEM and PURCHASE respectively, indicated by solid lines with hollow triangle heads. PURCHASE_ITEM and PURCHASE are connected by a solid line with hollow triangle heads at both ends, indicating a bidirectional association.	3
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